# ActualHonorsProject

November 8, 2026

[9]:
```
import pandas as pd
import numpy as np
from sklearn import preprocessing
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import seaborn as sns
```

[10]:
```
housing = pd.read_csv("Housing.csv")
```

[11]:
```
df = pd.DataFrame(housing)
```

[12]:
```
len(housing)
```

[12]: 4600

[13]:
```
df.head()
```

[13]:

| | date | price | bedrooms | bathrooms | sqft_living | sqft_lot | floors | waterfront | view | condition | sqft_above | sqft_basement | yr_built | yr_renovated | street | city | statezip | price_per_sqft |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2014-05-02 | 313000.0 | 3.0 | 1.50 | 1340 | 7912 | 1.5 | 0 | 0 | 3 | 1340 | 0 | 1955 | 2005 | 18810 Densmore Ave N | Shoreline | WA 98133 | 233.58 |
| 1 | 2014-05-02 | 2384000.0 | 5.0 | 2.50 | 3650 | 9050 | 2.0 | 0 | 4 | 5 | 3370 | 280 | 1921 | 0 | 709 W Blaine St | Seattle | WA 98119 | 653.15 |
| 2 | 2014-05-02 | 342000.0 | 3.0 | 2.00 | 1930 | 11947 | 1.0 | 0 | 0 | 4 | 1930 | 0 | 1966 | 0 | 26206-26214 143rd Ave SE | Kent | WA 98042 | 177.20 |
| 3 | 2014-05-02 | 420000.0 | 3.0 | 2.25 | 2000 | 8030 | 1.0 | 0 | 0 | 4 | 1000 | 1000 | 1963 | 0 | 857 170th Pl NE | Bellevue | WA 98008 | 210.00 |
| 4 | 2014-05-02 | 550000.0 | 4.0 | 2.50 | 1940 | 10500 | 1.0 | 0 | 0 | 4 | 1140 | 800 | 1976 | 0 | 9105 170th Ave NE | Redmond | WA 98052 | 283.71 |

```
4          1992         9105 170th Ave NE    Redmond  WA 98052          283.51
```

[14]:
```
df.tail()
```

[14]:
```
            date      price  bedrooms  bathrooms  sqft_living  sqft_lot  \
4595  2014-07-09  308166.67       3.0       1.75         1510      6360
4596  2014-07-09  534333.33       3.0       2.50         1460      7573
4597  2014-07-09  416904.17       3.0       2.50         3010      7014
4598  2014-07-10  203400.00       4.0       2.00         2090      6630
4599  2014-07-10  220600.00       3.0       2.50         1490      8102

      floors  waterfront  view  condition  sqft_above  sqft_basement  \
4595     1.0           0     0          4        1510              0
4596     2.0           0     0          3        1460              0
4597     2.0           0     0          3        3010              0
4598     1.0           0     0          3        1070           1020
4599     2.0           0     0          4        1490              0

      yr_built  yr_renovated             street       city  statezip  \
4595      1954          1979     501 N 143rd St    Seattle  WA 98133
4596      1983          2009    14855 SE 10th Pl   Bellevue  WA 98007
4597      2009             0    759 Ilwaco Pl NE     Renton  WA 98059
4598      1974             0   5148 S Creston St    Seattle  WA 98178
4599      1990             0  18717 SE 258th St  Covington  WA 98042

      price_per_sqft
4595          204.08
4596          365.98
4597          138.51
4598           97.32
4599          148.05
```

[15]:
```
df.columns
```

[15]:
```
Index(['date', 'price', 'bedrooms', 'bathrooms', 'sqft_living', 'sqft_lot',
       'floors', 'waterfront', 'view', 'condition', 'sqft_above',
       'sqft_basement', 'yr_built', 'yr_renovated', 'street', 'city',
       'statezip', 'price_per_sqft'],
      dtype='object')
```

### Exploratory Data Analysis and Visualizations

[16]:
```
df['price'].describe()
```

[16]:
```
count    4.600000e+03
mean     5.519630e+05
std      5.638347e+05
```

```
min       0.000000e+00
25%       3.228750e+05
50%       4.609435e+05
75%       6.549625e+05
max       2.659000e+07
Name: price, dtype: float64
```

[17]:
```
df.loc[df['price']==0]
```

[17]:

| | date | price | bedrooms | bathrooms | sqft_living | sqft_lot | floors | \ |
|---|---|---|---|---|---|---|---|---|
| 4354 | 2014-05-05 | 0.0 | 3.0 | 1.75 | 1490 | 10125 | 1.0 | |
| 4356 | 2014-05-05 | 0.0 | 4.0 | 2.75 | 2600 | 5390 | 1.0 | |
| 4357 | 2014-05-05 | 0.0 | 6.0 | 2.75 | 3200 | 9200 | 1.0 | |
| 4358 | 2014-05-06 | 0.0 | 5.0 | 3.50 | 3480 | 36615 | 2.0 | |
| 4361 | 2014-05-07 | 0.0 | 5.0 | 1.50 | 1500 | 7112 | 1.0 | |
| 4362 | 2014-05-07 | 0.0 | 4.0 | 4.00 | 3680 | 18804 | 2.0 | |
| 4374 | 2014-05-09 | 0.0 | 2.0 | 2.50 | 2200 | 188200 | 1.0 | |
| 4376 | 2014-05-09 | 0.0 | 4.0 | 2.25 | 2170 | 10500 | 1.0 | |
| 4382 | 2014-05-12 | 0.0 | 5.0 | 4.50 | 4630 | 6324 | 2.0 | |
| 4383 | 2014-05-13 | 0.0 | 5.0 | 4.00 | 4430 | 9000 | 2.0 | |
| 4385 | 2014-05-13 | 0.0 | 4.0 | 4.50 | 5030 | 11023 | 2.0 | |
| 4386 | 2014-05-13 | 0.0 | 4.0 | 1.50 | 2180 | 22870 | 1.0 | |
| 4389 | 2014-05-15 | 0.0 | 4.0 | 3.50 | 4210 | 10308 | 2.0 | |
| 4394 | 2014-05-16 | 0.0 | 5.0 | 3.25 | 3690 | 12353 | 2.0 | |
| 4405 | 2014-05-20 | 0.0 | 4.0 | 3.75 | 3300 | 4545 | 1.5 | |
| 4408 | 2014-05-21 | 0.0 | 5.0 | 2.25 | 2880 | 11965 | 2.0 | |
| 4411 | 2014-05-22 | 0.0 | 5.0 | 2.25 | 2000 | 7900 | 1.0 | |
| 4412 | 2014-05-22 | 0.0 | 3.0 | 3.00 | 1860 | 7440 | 1.0 | |
| 4413 | 2014-05-22 | 0.0 | 4.0 | 3.00 | 1990 | 6180 | 2.0 | |
| 4420 | 2014-05-27 | 0.0 | 4.0 | 1.00 | 1360 | 13372 | 1.0 | |
| 4442 | 2014-06-02 | 0.0 | 1.0 | 1.00 | 720 | 6000 | 1.0 | |
| 4448 | 2014-06-03 | 0.0 | 5.0 | 2.75 | 2740 | 5616 | 1.5 | |
| 4453 | 2014-06-03 | 0.0 | 3.0 | 1.00 | 1300 | 6710 | 1.0 | |
| 4454 | 2014-06-03 | 0.0 | 5.0 | 2.50 | 2090 | 4698 | 2.0 | |
| 4472 | 2014-06-09 | 0.0 | 4.0 | 3.75 | 4060 | 19290 | 2.0 | |
| 4478 | 2014-06-11 | 0.0 | 5.0 | 2.75 | 2910 | 53898 | 1.0 | |
| 4479 | 2014-06-11 | 0.0 | 5.0 | 2.00 | 1910 | 7200 | 1.0 | |
| 4480 | 2014-06-11 | 0.0 | 3.0 | 2.50 | 2880 | 13500 | 1.0 | |
| 4481 | 2014-06-11 | 0.0 | 5.0 | 2.75 | 3240 | 6863 | 2.0 | |
| 4487 | 2014-06-12 | 0.0 | 4.0 | 1.00 | 2080 | 3500 | 1.5 | |
| 4499 | 2014-06-17 | 0.0 | 5.0 | 3.75 | 3870 | 8225 | 2.0 | |
| 4507 | 2014-06-18 | 0.0 | 4.0 | 1.50 | 2310 | 68824 | 2.0 | |
| 4509 | 2014-06-18 | 0.0 | 6.0 | 3.00 | 3020 | 13783 | 2.0 | |
| 4520 | 2014-06-20 | 0.0 | 4.0 | 2.50 | 1960 | 11600 | 1.0 | |
| 4521 | 2014-06-20 | 0.0 | 4.0 | 1.00 | 1810 | 7500 | 1.0 | |
| 4522 | 2014-06-22 | 0.0 | 2.0 | 2.25 | 1490 | 6770 | 1.5 | |
| 4523 | 2014-06-23 | 0.0 | 3.0 | 4.50 | 5230 | 17826 | 2.0 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 4528 | 2014-06-24 | 0.0 | 4.0 | 5.00 | 4550 | 18641 | 1.0 |
| 4534 | 2014-06-24 | 0.0 | 3.0 | 2.75 | 1310 | 7300 | 1.0 |
| 4542 | 2014-06-25 | 0.0 | 5.0 | 3.50 | 2640 | 6895 | 2.0 |
| 4552 | 2014-06-26 | 0.0 | 4.0 | 2.00 | 2100 | 4857 | 2.0 |
| 4554 | 2014-06-27 | 0.0 | 2.0 | 1.00 | 810 | 8424 | 1.0 |
| 4555 | 2014-06-27 | 0.0 | 2.0 | 1.50 | 1520 | 8040 | 1.0 |
| 4558 | 2014-06-28 | 0.0 | 4.0 | 4.25 | 3500 | 8750 | 1.0 |
| 4563 | 2014-07-01 | 0.0 | 2.0 | 2.25 | 2130 | 4920 | 1.5 |
| 4567 | 2014-07-02 | 0.0 | 4.0 | 2.50 | 4080 | 18362 | 2.0 |
| 4574 | 2014-07-02 | 0.0 | 3.0 | 1.00 | 1520 | 9030 | 1.0 |
| 4575 | 2014-07-02 | 0.0 | 5.0 | 6.25 | 8020 | 21738 | 2.0 |
| 4588 | 2014-07-08 | 0.0 | 4.0 | 2.25 | 2890 | 18226 | 3.0 |

| | waterfront | view | condition | sqft_above | sqft_basement | yr_built | \ |
|---|---|---|---|---|---|---|---|
| 4354 | 0 | 0 | 4 | 1490 | 0 | 1962 | |
| 4356 | 0 | 0 | 4 | 1300 | 1300 | 1960 | |
| 4357 | 0 | 2 | 4 | 1600 | 1600 | 1953 | |
| 4358 | 0 | 0 | 4 | 2490 | 990 | 1983 | |
| 4361 | 0 | 0 | 5 | 760 | 740 | 1920 | |
| 4362 | 0 | 0 | 3 | 3680 | 0 | 1990 | |
| 4374 | 0 | 3 | 3 | 2200 | 0 | 2007 | |
| 4376 | 0 | 2 | 4 | 1270 | 900 | 1960 | |
| 4382 | 0 | 0 | 3 | 3210 | 1420 | 2006 | |
| 4383 | 0 | 0 | 3 | 4430 | 0 | 2013 | |
| 4385 | 0 | 2 | 3 | 3250 | 1780 | 2008 | |
| 4386 | 0 | 0 | 4 | 1280 | 900 | 1954 | |
| 4389 | 0 | 0 | 3 | 4210 | 0 | 2006 | |
| 4394 | 0 | 0 | 5 | 3690 | 0 | 1977 | |
| 4405 | 0 | 4 | 3 | 2600 | 700 | 1926 | |
| 4408 | 0 | 0 | 4 | 2880 | 0 | 1990 | |
| 4411 | 0 | 0 | 4 | 1300 | 700 | 1986 | |
| 4412 | 0 | 0 | 5 | 1040 | 820 | 1954 | |
| 4413 | 0 | 0 | 3 | 1990 | 0 | 1990 | |
| 4420 | 0 | 0 | 3 | 1360 | 0 | 1955 | |
| 4442 | 0 | 0 | 3 | 720 | 0 | 1940 | |
| 4448 | 0 | 0 | 5 | 1670 | 1070 | 1925 | |
| 4453 | 0 | 0 | 4 | 1300 | 0 | 1952 | |
| 4454 | 0 | 0 | 3 | 2090 | 0 | 1998 | |
| 4472 | 0 | 0 | 3 | 4060 | 0 | 2002 | |
| 4478 | 0 | 0 | 5 | 1510 | 1400 | 1979 | |
| 4479 | 0 | 0 | 4 | 1110 | 800 | 1951 | |
| 4480 | 0 | 4 | 5 | 1520 | 1360 | 1950 | |
| 4481 | 0 | 0 | 3 | 3240 | 0 | 2013 | |
| 4487 | 0 | 0 | 5 | 1260 | 820 | 1926 | |
| 4499 | 0 | 0 | 3 | 3870 | 0 | 1998 | |
| 4507 | 0 | 0 | 4 | 2310 | 0 | 1968 | |
| 4509 | 0 | 0 | 3 | 3020 | 0 | 1952 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| 4520 | 0 | 0 | 5 | 980 | 980 | 1931 |
| 4521 | 0 | 0 | 2 | 1410 | 400 | 1959 |
| 4522 | 0 | 0 | 3 | 1490 | 0 | 1926 |
| 4523 | 1 | 4 | 3 | 3740 | 1490 | 2005 |
| 4528 | 1 | 4 | 3 | 2600 | 1950 | 2002 |
| 4534 | 0 | 0 | 3 | 1310 | 0 | 1957 |
| 4542 | 0 | 0 | 3 | 2640 | 0 | 2001 |
| 4552 | 0 | 0 | 3 | 2100 | 0 | 1965 |
| 4554 | 0 | 0 | 4 | 810 | 0 | 1959 |
| 4555 | 0 | 0 | 5 | 1520 | 0 | 1951 |
| 4558 | 0 | 4 | 5 | 2140 | 1360 | 1951 |
| 4563 | 0 | 4 | 4 | 1530 | 600 | 1941 |
| 4567 | 0 | 2 | 4 | 4080 | 0 | 1983 |
| 4574 | 0 | 0 | 3 | 1520 | 0 | 1956 |
| 4575 | 0 | 0 | 3 | 8020 | 0 | 2001 |
| 4588 | 1 | 4 | 3 | 2890 | 0 | 1984 |

| | yr_renovated | street | city | statezip | \ |
|---|---|---|---|---|---|
| 4354 | 0 | 3911 S 328th St | Federal Way | WA 98001 | |
| 4356 | 2001 | 2120 31st Ave W | Seattle | WA 98199 | |
| 4357 | 1983 | 12271 Marine View Dr SW | Burien | WA 98146 | |
| 4358 | 0 | 21809 SE 38th Pl | Issaquah | WA 98075 | |
| 4361 | 0 | 14901-14999 12th Ave SW | Burien | WA 98166 | |
| 4362 | 2009 | 1223-1237 244th Ave NE | Sammamish | WA 98074 | |
| 4374 | 0 | 39612 254th Ave SE | Enumclaw | WA 98022 | |
| 4376 | 2001 | 216 SW 183rd St | Normandy Park | WA 98166 | |
| 4382 | 0 | 6925 Oakmont Ave SE | Snoqualmie | WA 98065 | |
| 4383 | 1923 | 9235 NE 5th St | Bellevue | WA 98004 | |
| 4385 | 0 | 4140 Boulevard Pl | Mercer Island | WA 98040 | |
| 4386 | 1975 | 31603 E Lake Morton Dr SE | Kent | WA 98042 | |
| 4389 | 0 | 2234 167th Ave SE | Bellevue | WA 98008 | |
| 4394 | 0 | 19055 35th Ave NE | Lake Forest Park | WA 98155 | |
| 4405 | 1999 | 3665 50th Ave NE | Seattle | WA 98105 | |
| 4408 | 0 | 25437 163rd Pl SE | Covington | WA 98042 | |
| 4411 | 0 | 3202 S 194th St | SeaTac | WA 98188 | |
| 4412 | 0 | 10744 62nd Ave S | Seattle | WA 98178 | |
| 4413 | 2009 | 32706 20th Ave SW | Federal Way | WA 98023 | |
| 4420 | 2005 | 18423 61st Pl NE | Kenmore | WA 98028 | |
| 4442 | 1996 | 1236 S Cloverdale St | Seattle | WA 98108 | |
| 4448 | 0 | 1013 NE 80th St | Seattle | WA 98115 | |
| 4453 | 0 | 2760 72nd Ave SE | Mercer Island | WA 98040 | |
| 4454 | 2006 | 27622 237th Pl SE | Maple Valley | WA 98038 | |
| 4472 | 0 | 21418 SE 5th Pl | Sammamish | WA 98074 | |
| 4478 | 0 | 13505 208th Ave NE | Woodinville | WA 98077 | |
| 4479 | 1999 | 11620-11698 57th Ave S | Seattle | WA 98178 | |
| 4480 | 0 | 9243 NE 20th St | Clyde Hill | WA 98004 | |
| 4481 | 1923 | 1301-1303 Monterey Ave NE | Renton | WA 98056 | |

| | | | | | |
|---|---|---|---|---|---|
| 4487 | 0 | 6506 40th Ave SW | Seattle | WA | 98136 |
| 4499 | 2006 | 101-127 247th Ave SE | Sammamish | WA | 98074 |
| 4507 | 0 | 29656 232nd Ave SE | Black Diamond | WA | 98010 |
| 4509 | 2002 | 4115 85th Ave SE | Mercer Island | WA | 98040 |
| 4520 | 0 | 506 21st St SE | Auburn | WA | 98002 |
| 4521 | 0 | 12231 Occidental Ave S | Seattle | WA | 98168 |
| 4522 | 2003 | 4921 28th Ave S | Seattle | WA | 98108 |
| 4523 | 0 | 7455 W Mercer Way | Mercer Island | WA | 98040 |
| 4528 | 0 | 425 E Lake Sammamish Pkwy SE | Sammamish | WA | 98074 |
| 4534 | 2000 | 16232 SE 10th St | Bellevue | WA | 98008 |
| 4542 | 0 | 34529 SE Jay Ct | Snoqualmie | WA | 98065 |
| 4552 | 1984 | 4500 NE 171st St | Lake Forest Park | WA | 98155 |
| 4554 | 0 | 30401-30499 8th Ave SW | Federal Way | WA | 98023 |
| 4555 | 0 | 11533 22nd Ave NE | Seattle | WA | 98125 |
| 4558 | 0 | 12725 8th Ave NW | Seattle | WA | 98177 |
| 4563 | 1998 | 3428 60th Ave SW | Seattle | WA | 98116 |
| 4567 | 0 | 2710 95th Ave NE | Clyde Hill | WA | 98004 |
| 4574 | 2001 | 2533 155th Pl SE | Bellevue | WA | 98007 |
| 4575 | 0 | 2 Crescent Key | Bellevue | WA | 98006 |
| 4588 | 0 | 3227-3399 Mountain View Ave N | Renton | WA | 98056 |

| | price_per_sqft |
|---|---|
| 4354 | 0.0 |
| 4356 | 0.0 |
| 4357 | 0.0 |
| 4358 | 0.0 |
| 4361 | 0.0 |
| 4362 | 0.0 |
| 4374 | 0.0 |
| 4376 | 0.0 |
| 4382 | 0.0 |
| 4383 | 0.0 |
| 4385 | 0.0 |
| 4386 | 0.0 |
| 4389 | 0.0 |
| 4394 | 0.0 |
| 4405 | 0.0 |
| 4408 | 0.0 |
| 4411 | 0.0 |
| 4412 | 0.0 |
| 4413 | 0.0 |
| 4420 | 0.0 |
| 4442 | 0.0 |
| 4448 | 0.0 |
| 4453 | 0.0 |
| 4454 | 0.0 |
| 4472 | 0.0 |

```
4478            0.0
4479            0.0
4480            0.0
4481            0.0
4487            0.0
4499            0.0
4507            0.0
4509            0.0
4520            0.0
4521            0.0
4522            0.0
4523            0.0
4528            0.0
4534            0.0
4542            0.0
4552            0.0
4554            0.0
4555            0.0
4558            0.0
4563            0.0
4567            0.0
4574            0.0
4575            0.0
4588            0.0
```

What to do about houses where price is 0? For now, going to leave as 0.

[18]:
```
df.isna().sum()
```

[18]:
```
date              0
price             0
bedrooms          0
bathrooms         0
sqft_living       0
sqft_lot          0
floors            0
waterfront        0
view              0
condition         0
sqft_above        0
sqft_basement     0
yr_built          0
yr_renovated      0
street            0
city              0
statezip          0
price_per_sqft    0
```

```
dtype: int64
```

[19]:
```
df.loc[df['price']==2.659000e+07]
```

[19]:
```
            date       price  bedrooms  bathrooms  sqft_living  sqft_lot  \
4350  2014-07-03  26590000.0       3.0        2.0         1180      7793

      floors  waterfront  view  condition  sqft_above  sqft_basement  \
4350     1.0           0     0          4        1180              0

      yr_built  yr_renovated             street  city  statezip  \
4350      1992             0  12005 SE 219th Ct  Kent  WA 98031

      price_per_sqft
4350         22533.9
```

want to find average price per square foot for houses with no waterfront vs with waterfront

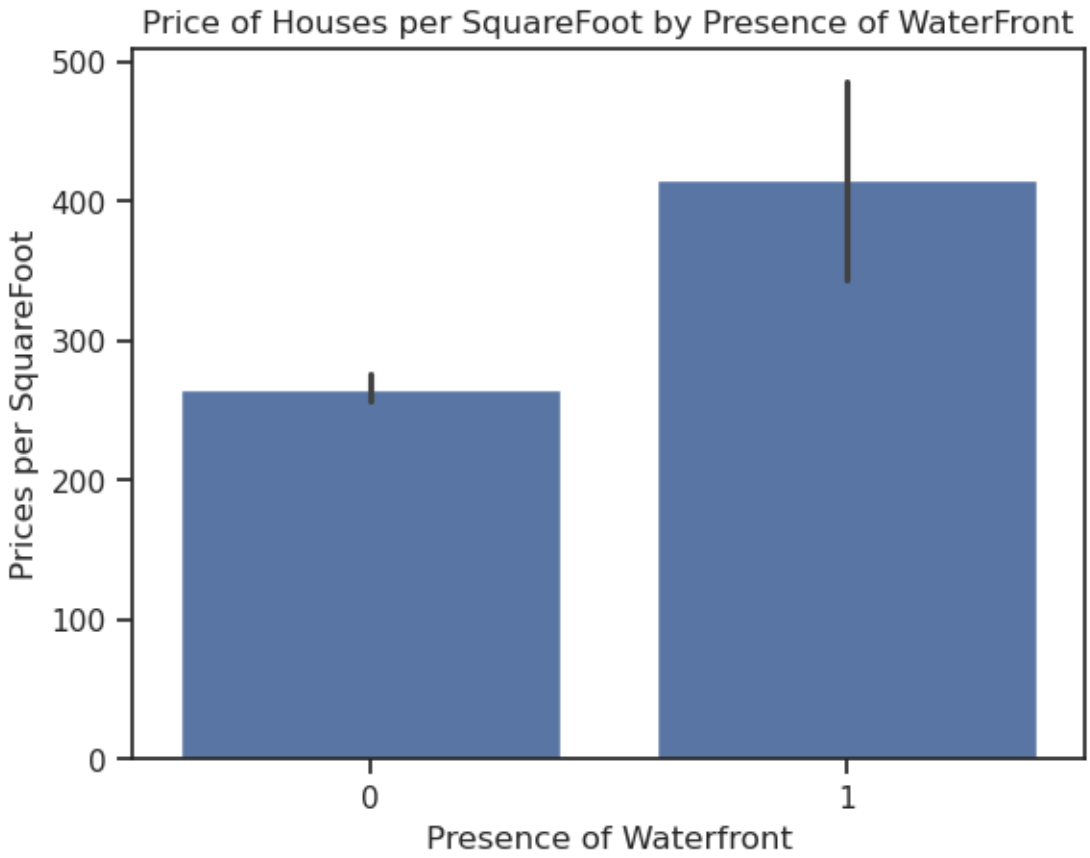
[20]:
```
no_water = df.loc[df["waterfront"]==0]
no_water["price_per_sqft"].mean()
```

[20]:
```
264.7979132910006
```

[21]:
```
water = df.loc[df["waterfront"]==1]
water["price_per_sqft"].mean()
```

[21]:
```
415.10575757575765
```

[22]:
```
sns.set_theme(style="ticks")
sns.barplot(y = "price_per_sqft", x = "waterfront", data = housing)
plt.title('Price of Houses per SquareFoot by Presence of WaterFront')
plt.xlabel('Presence of Waterfront')
plt.ylabel('Prices per SquareFoot')
plt.show()
```




[23]:
```
df["price"].corr(df["waterfront"])
```

[23]:
```
0.13564832245964026
```

Very low correlation

[24]:
```
sns.set_theme(style="ticks")
sns.barplot(y = "price_per_sqft", x = "bedrooms", data = housing)
plt.title('Price of Houses per SquareFoot by Number of Bedrooms')
plt.xlabel('Number of Bedrooms')
plt.ylabel('Prices per SquareFoot')
plt.show()
```




[25]:
```
df["price"].corr(df["bedrooms"])
```

[25]:
```
0.20033628938558576
```

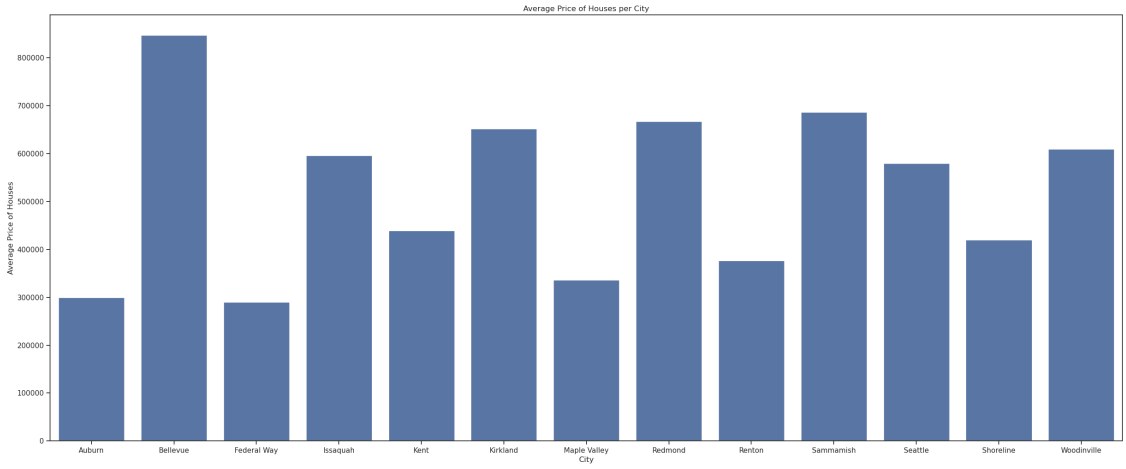
[26]:
```
cityinfo = df.groupby('city', as_index = False).agg(average_price =␣
↪('price','mean'), average_pricepersqft = ('price_per_sqft','mean'),␣
↪num_of_houses = ('price','count'))
cityinfo.sort_values(by='num_of_houses')
cityINFO=cityinfo.loc[cityinfo["num_of_houses"]>86]
cityINFO
```

[26]:

| | city | average_price | average_pricepersqft | num_of_houses |
|---|---|---|---|---|
| 1 | Auburn | 299340.442784 | 151.345511 | 176 |
| 3 | Bellevue | 847180.662972 | 318.094720 | 286 |
| 14 | Federal Way | 289887.703986 | 145.379257 | 148 |
| 16 | Issaquah | 596163.747487 | 250.167754 | 187 |
| 18 | Kent | 439492.444703 | 275.079405 | 185 |
| 19 | Kirkland | 651583.591925 | 288.279144 | 187 |
| 21 | Maple Valley | 336474.911771 | 163.530417 | 96 |
| 31 | Redmond | 667649.534766 | 278.734766 | 235 |

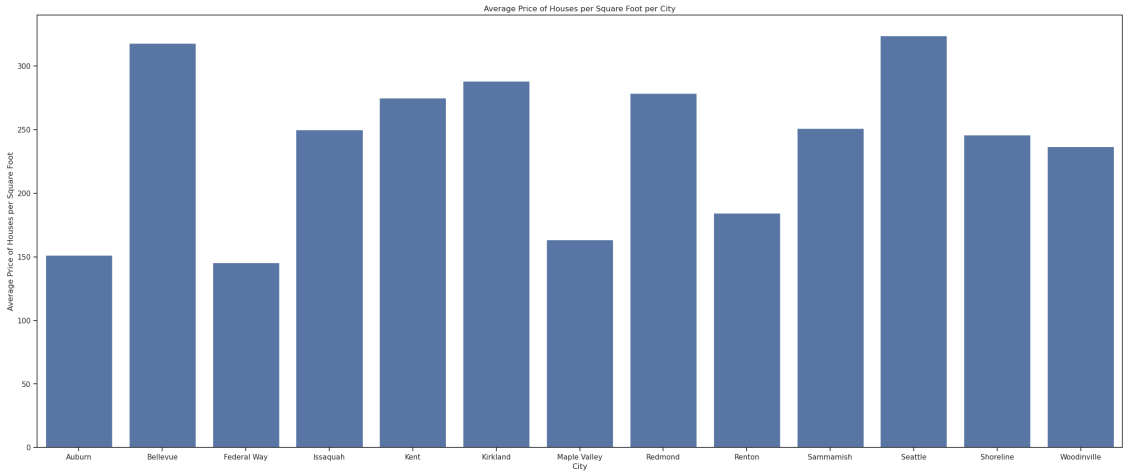
| | | | | |
|---|---|---|---|---|
| 32 | Renton | 377040.966382 | 184.641365 | 293 |
| 33 | Sammamish | 686917.583486 | 251.249771 | 175 |
| 35 | Seattle | 579837.467279 | 323.927184 | 1573 |
| 36 | Shoreline | 420392.364065 | 245.906179 | 123 |
| 42 | Woodinville | 609565.021739 | 236.768696 | 115 |

[27]:
```
plt.figure(figsize=(30,12))
sns.set_theme(style="ticks")
sns.barplot(y = "average_price", x = "city", data = cityINFO)
plt.title('Average Price of Houses per City')
plt.xlabel('City')
plt.ylabel('Average Price of Houses')
plt.show()
```



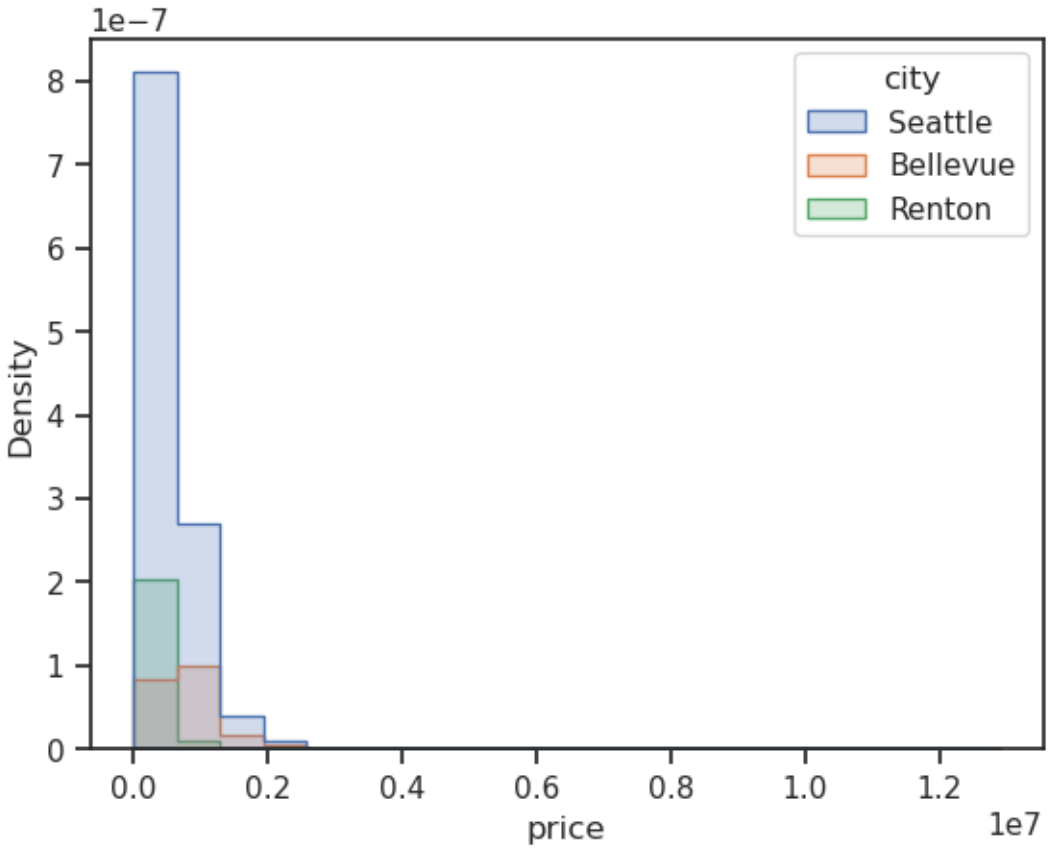

Explanation of the code: created a new dataframe for the city, number of cities, average price of houses and avergae price of houses by square feet. Many cities only had a small number of houses listed in the original data set, so out of the 44 cities I took the top 13 cities with the most houses. Then I did a bar graph of these cities with the average price. Bellevue seems to have the highest prices, and Auburn seems to be tied with Federal Way for the lowest house prices.

[28]:
```
plt.figure(figsize=(30,12))
sns.set_theme(style="ticks")
sns.barplot(y = "average_pricepersqft", x = "city", data = cityINFO)
plt.title('Average Price of Houses per Square Foot per City')
plt.xlabel('City')
plt.ylabel('Average Price of Houses per Square Foot')
plt.show()
```




[29]:
```
cityinfo.sort_values(by='num_of_houses', ascending = False).head(3)
housingnew = housing.loc[housing['city'].isin(['Seattle','Renton','Bellevue'])]
housingnew
sns.histplot(data = housingnew, x = "price", stat = "density",
             bins = 20, hue = "city", element = "step")
```

[29]: <Axes: xlabel='price', ylabel='Density'>




[30]:
```
areainfo = df.groupby('statezip', as_index = False).agg(average_price =␣
 ↪('price','mean'), num_of_houses = ('price','count'))
areainfo.sort_values(by='num_of_houses')
```

[30]:

| | statezip | average_price | num_of_houses |
|---|---|---|---|
| 39 | WA 98068 | 525000.000000 | 1 |
| 76 | WA 98354 | 285000.000000 | 2 |
| 29 | WA 98050 | 562450.000000 | 2 |
| 75 | WA 98288 | 233000.000000 | 3 |
| 28 | WA 98047 | 225233.333333 | 6 |
| .. | … | … | … |
| 5 | WA 98006 | 793052.634091 | 110 |
| 54 | WA 98115 | 595086.468154 | 130 |
| 56 | WA 98117 | 563904.759697 | 132 |
| 31 | WA 98052 | 637555.719333 | 135 |
| 47 | WA 98103 | 578663.061757 | 148 |

[77 rows x 3 columns]

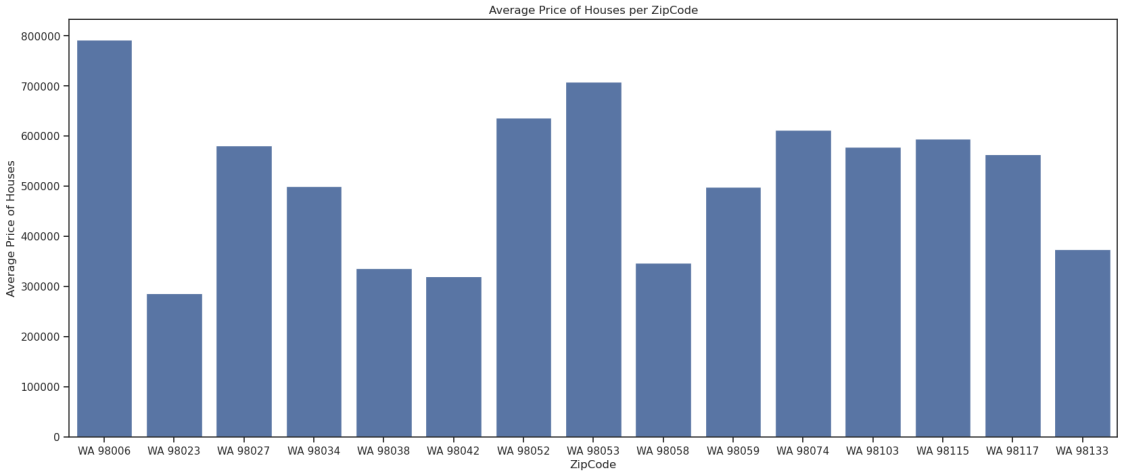
[31]:
```
areaINFO=areainfo.loc[areainfo["num_of_houses"]>89]
areaINFO.sort_values(by='num_of_houses')
```

[31]:

| | statezip | average_price | num_of_houses |
|---|---|---|---|
| 62 | WA 98133 | 374433.233011 | 93 |
| 13 | WA 98023 | 287306.412526 | 95 |
| 36 | WA 98058 | 347928.795895 | 95 |
| 15 | WA 98027 | 581749.989474 | 95 |
| 23 | WA 98038 | 336474.911771 | 96 |
| 42 | WA 98074 | 613305.538061 | 98 |
| 32 | WA 98053 | 708179.781224 | 98 |
| 22 | WA 98034 | 501118.971616 | 99 |
| 26 | WA 98042 | 320243.968500 | 100 |
| 37 | WA 98059 | 499455.770189 | 106 |
| 5 | WA 98006 | 793052.634091 | 110 |
| 54 | WA 98115 | 595086.468154 | 130 |
| 56 | WA 98117 | 563904.759697 | 132 |
| 31 | WA 98052 | 637555.719333 | 135 |
| 47 | WA 98103 | 578663.061757 | 148 |

These are the 15 areacodes with the highest number of houses (more data)

[32]:
```
plt.figure(figsize=(20,8))
sns.set_theme(style="ticks")
sns.barplot(y = "average_price", x = "statezip", data = areaINFO)
plt.title('Average Price of Houses per ZipCode')
plt.xlabel('ZipCode')
plt.ylabel('Average Price of Houses')
plt.show()
```




is 98006 located in Bellevue?

[33]:
```
HOUSING = housing.loc[housing['statezip'] == "WA 98006"]
```

[34]:
```
HOUSING
```

[34]:

```
            date       price  bedrooms  bathrooms  sqft_living  sqft_lot  \
21    2014-05-02   435000.00       4.0       1.00         1450      8800
55    2014-05-02   430000.00       4.0       1.50         1920     10000
95    2014-05-05   541125.00       5.0       2.75         2740      8426
118   2014-05-05  1030000.00       5.0       2.75         3190     16920
171   2014-05-06   736000.00       4.0       2.50         2290     12047
…            …          …       …         …            …        …
4476  2014-06-10   812650.00       4.0       2.50         3700     21755
4500  2014-06-17   540000.00       3.0       2.75         2750     18029
4575  2014-07-02        0.00       5.0       6.25         8020     21738
4583  2014-07-07   282766.67       4.0       2.00         1700      8640
4584  2014-07-07   486445.83       3.0       1.75         1880     10032

      floors  waterfront  view  condition  sqft_above  sqft_basement  \
21       1.0           0     0          4        1450              0
55       1.0           0     0          4        1070            850
95       1.0           0     0          4        1370           1370
118      1.0           0     3          3        1690           1500
171      2.0           0     0          4        2290              0
…       …          …    …         …          …             …
4476     1.0           0     4          3        2620           1080
4500     1.0           0     2          5        1810            940
4575     2.0           0     0          3        8020              0
4583     1.0           0     0          3         850            850
4584     1.0           0     0          4        1880              0

      yr_built  yr_renovated             street      city  statezip  \
21        1954          1979  3922 154th Ave SE  Bellevue  WA 98006
55        1954          1979  4071 156th Ave SE  Bellevue  WA 98006
95        1960          2001   5211 116th Pl SE  Bellevue  WA 98006
118       1976             0  4529 135th Ave SE  Bellevue  WA 98006
171       1988             0   17435 SE 47th St  Bellevue  WA 98006
…        …           …                 …       …       …
4476      1988          2000   4462 175th Pl SE  Bellevue  WA 98006
4500      1978             0   4708 154th Pl SE  Bellevue  WA 98006
4575      2001             0     2 Crescent Key  Bellevue  WA 98006
4583      1955          2010  3706 140th Ave SE  Bellevue  WA 98006
4584      1984             0   13345 SE 55th Pl  Bellevue  WA 98006

      price_per_sqft
21            300.00
55            223.96
95            197.49
```

```
118           322.88
171           321.40
…               …
4476          219.64
4500          196.36
4575            0.00
4583          166.33
4584          258.75

[110 rows x 18 columns]
```

Yes, 98006, the most expensive zip code is located in Bellevue, which also had the most expensive average price for houses.

[35]:
```
conditioninfo = df.groupby('condition', as_index = False).agg(average_price =␣
↪('price','mean'))
```

[36]:
```
plt.figure(figsize=(7,5))
sns.set_theme(style="ticks")
sns.barplot(y = "average_price", x = "condition", data = conditioninfo)
plt.title('Average Price of Houses per Condition')
plt.xlabel('Condition')
plt.ylabel('Average Price of Houses')
plt.show()
```

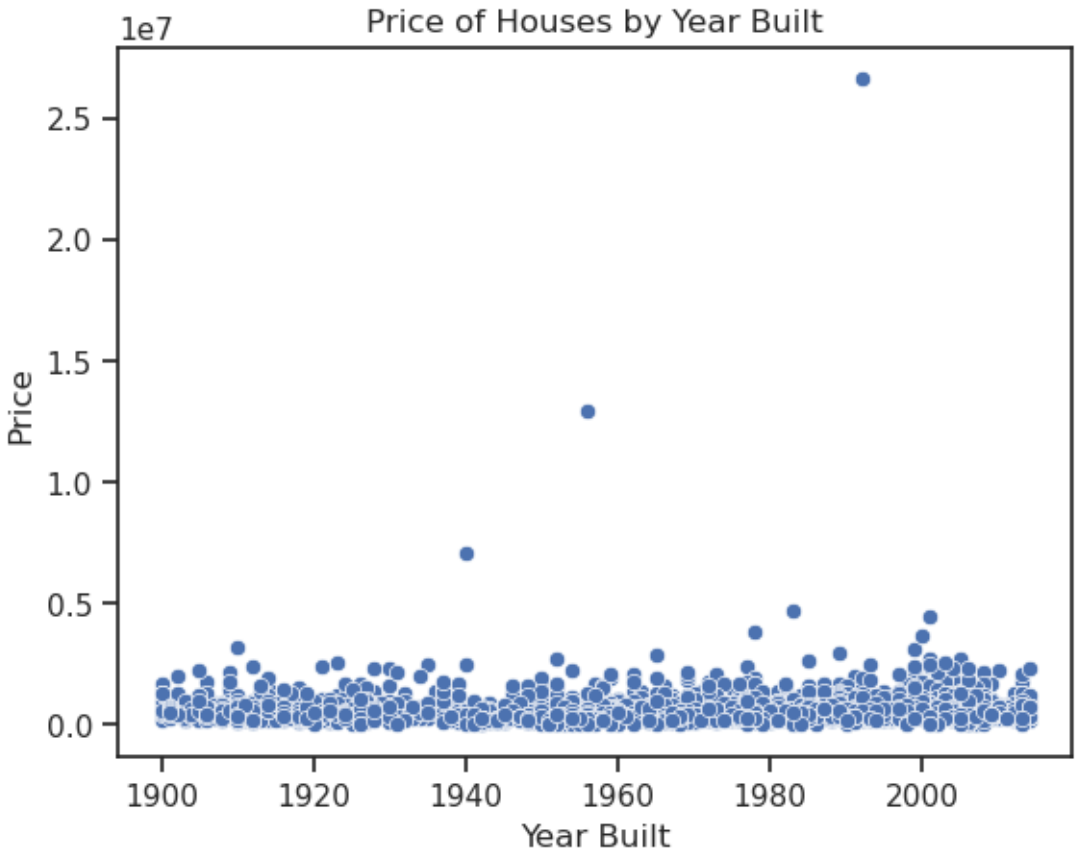


[37]:
```
conditioninfo["condition"].corr(conditioninfo["average_price"])
```

[37]:
```
0.936790453176485
```

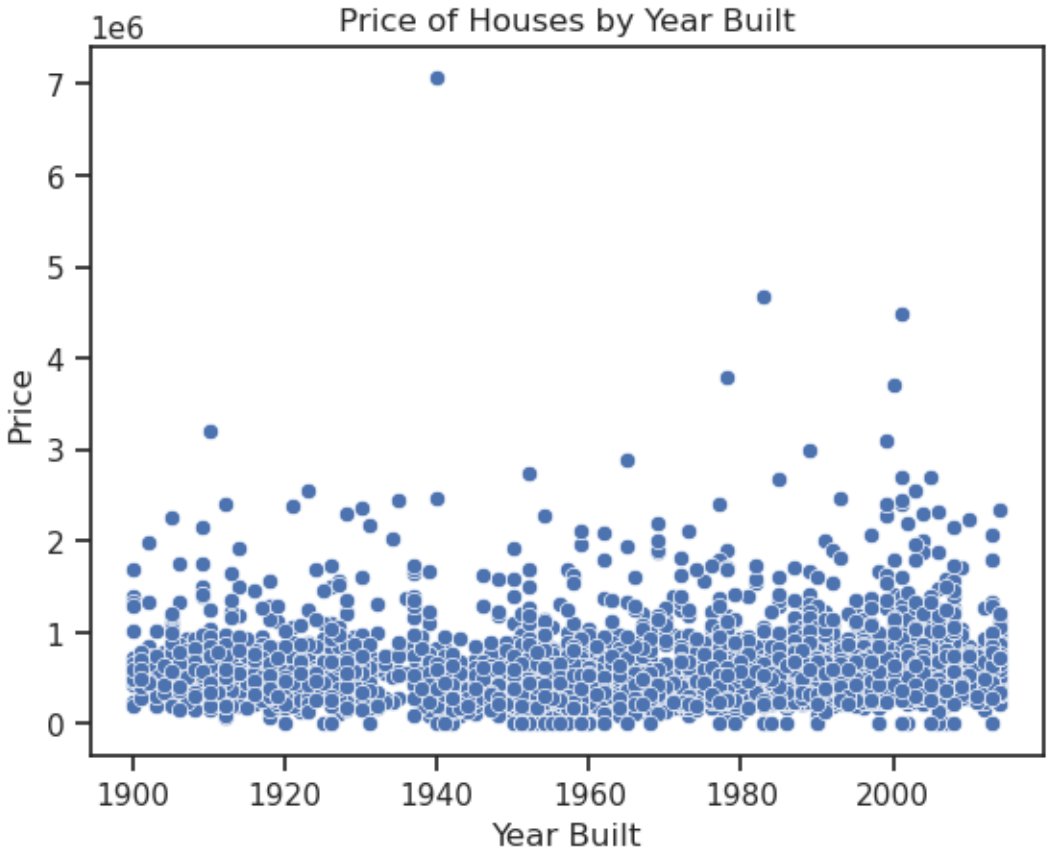
VERY HIGH correlation between condition and average price of houses

[38]:
```
sns.set_theme(style="ticks")
sns.scatterplot(y = "price", x = "yr_built", data = housing)
plt.title('Price of Houses by Year Built')
plt.xlabel('Year Built')
plt.ylabel('Price')
plt.show()
```



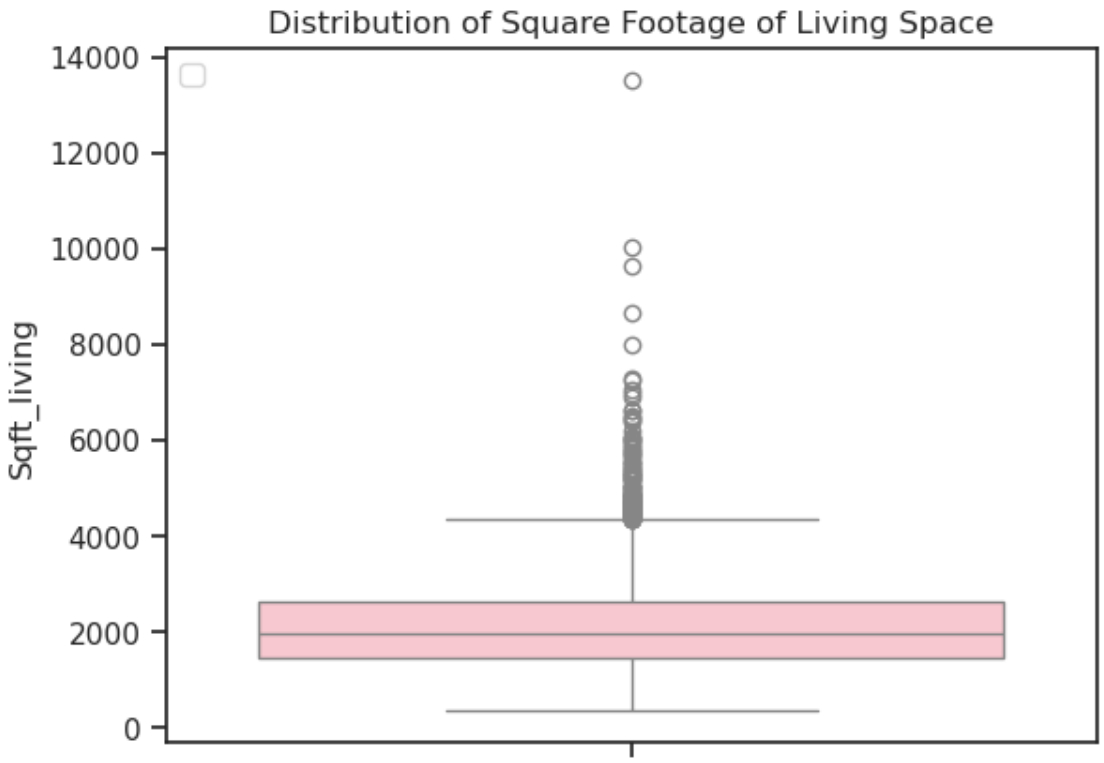

[39]:
```
YearINFO=housing.loc[housing["price"]<10000000]
sns.set_theme(style="ticks")
sns.scatterplot(y = "price", x = "yr_built", data = YearINFO)
plt.title('Price of Houses by Year Built')
plt.xlabel('Year Built')
```

```
plt.ylabel('Price')
plt.show()
```




[40]:
```
sns.boxplot(y = "sqft_living", data = housing, color = 'pink')
plt.legend(loc = "upper left")
plt.title('Distribution of Square Footage of Living Space')
plt.ylabel('Sqft_living')
plt.show()
```

```
/tmp/ipykernel_163/4262236960.py:2: UserWarning: No artists with labels found to
put in legend.  Note that artists whose label start with an underscore are
ignored when legend() is called with no argument.
  plt.legend(loc = "upper left")
```




[41]:
```
from scipy import stats
z_scores = stats.zscore(df['price'])
outliers = df[(z_scores>3) | (z_scores<-3)]
outliers
```

[41]:

| | date | price | bedrooms | bathrooms | sqft_living | sqft_lot | \ |
|---|---|---|---|---|---|---|---|
| 1 | 2014-05-02 | 2384000.00 | 5.0 | 2.50 | 3650 | 9050 | |
| 122 | 2014-05-05 | 2280000.00 | 7.0 | 8.00 | 13540 | 307752 | |
| 252 | 2014-05-07 | 3200000.00 | 7.0 | 4.50 | 6210 | 8856 | |
| 404 | 2014-05-09 | 2400000.00 | 3.0 | 2.25 | 3000 | 11665 | |
| 581 | 2014-05-13 | 2555000.00 | 4.0 | 2.50 | 5300 | 26211 | |
| 1132 | 2014-05-21 | 2700000.00 | 5.0 | 4.75 | 5305 | 8401 | |
| 1543 | 2014-05-29 | 2453500.00 | 4.0 | 3.50 | 4730 | 13586 | |
| 1567 | 2014-05-29 | 3100000.00 | 6.0 | 4.25 | 6980 | 15682 | |
| 1571 | 2014-05-29 | 2750000.00 | 4.0 | 3.25 | 4430 | 21000 | |
| 1600 | 2014-05-30 | 2400000.00 | 4.0 | 3.25 | 4140 | 20734 | |
| 1637 | 2014-05-30 | 3710000.00 | 4.0 | 3.50 | 5550 | 28078 | |
| 1858 | 2014-06-04 | 2400000.00 | 5.0 | 3.25 | 3410 | 9088 | |
| 2083 | 2014-06-08 | 2700000.00 | 4.0 | 3.25 | 4420 | 7850 | |
| 2286 | 2014-06-11 | 7062500.00 | 5.0 | 4.50 | 10040 | 37325 | |
| 2328 | 2014-06-11 | 2888000.00 | 5.0 | 6.25 | 8670 | 64033 | |
| 2364 | 2014-06-12 | 2250000.00 | 5.0 | 4.25 | 4860 | 9453 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| 2654 | 2014-06-17 | 4668000.00 | 5.0 | 6.75 | 9640 | 13068 |
| 2712 | 2014-06-17 | 2475000.00 | 3.0 | 3.25 | 4340 | 4947 |
| 2761 | 2014-06-18 | 4489000.00 | 4.0 | 3.00 | 6430 | 27517 |
| 2772 | 2014-06-18 | 3000000.00 | 4.0 | 4.25 | 4850 | 12445 |
| 2980 | 2014-06-20 | 2400000.00 | 6.0 | 4.50 | 5480 | 10800 |
| 3202 | 2014-06-24 | 2466350.00 | 5.0 | 4.75 | 6390 | 13180 |
| 3284 | 2014-06-25 | 2300000.00 | 4.0 | 3.25 | 4250 | 8570 |
| 3729 | 2014-07-01 | 3800000.00 | 5.0 | 5.50 | 7050 | 42840 |
| 3781 | 2014-07-01 | 2300000.00 | 4.0 | 4.75 | 3970 | 9778 |
| 3791 | 2014-07-01 | 2367000.00 | 3.0 | 2.25 | 3530 | 17450 |
| 3863 | 2014-07-02 | 2271150.00 | 4.0 | 3.25 | 4040 | 18916 |
| 3959 | 2014-07-03 | 2351956.00 | 4.0 | 4.25 | 5010 | 19412 |
| 4135 | 2014-07-08 | 2458000.00 | 4.0 | 5.25 | 6500 | 14986 |
| 4261 | 2014-07-09 | 2680000.00 | 5.0 | 3.00 | 4290 | 20445 |
| 4273 | 2014-07-09 | 2321000.00 | 5.0 | 4.75 | 5780 | 17004 |
| 4346 | 2014-06-23 | 12899000.00 | 3.0 | 2.50 | 2190 | 11394 |
| 4350 | 2014-07-03 | 26590000.00 | 3.0 | 2.00 | 1180 | 7793 |
| 4465 | 2014-06-05 | 2560498.33 | 3.0 | 2.50 | 1710 | 1664 |

| | floors | waterfront | view | condition | sqft_above | sqft_basement | \ |
|---|---|---|---|---|---|---|---|
| 1 | 2.0 | 0 | 4 | 5 | 3370 | 280 | |
| 122 | 3.0 | 0 | 4 | 3 | 9410 | 4130 | |
| 252 | 2.5 | 0 | 2 | 5 | 4760 | 1450 | |
| 404 | 1.5 | 1 | 4 | 3 | 3000 | 0 | |
| 581 | 2.0 | 1 | 2 | 2 | 4570 | 730 | |
| 1132 | 2.0 | 0 | 2 | 3 | 3745 | 1560 | |
| 1543 | 1.5 | 0 | 0 | 5 | 4270 | 460 | |
| 1567 | 3.0 | 0 | 4 | 4 | 5330 | 1650 | |
| 1571 | 2.0 | 0 | 0 | 3 | 4430 | 0 | |
| 1600 | 1.0 | 0 | 1 | 3 | 3300 | 840 | |
| 1637 | 2.0 | 0 | 2 | 4 | 3350 | 2200 | |
| 1858 | 2.0 | 0 | 3 | 3 | 2760 | 650 | |
| 2083 | 2.0 | 1 | 4 | 3 | 3150 | 1270 | |
| 2286 | 2.0 | 1 | 2 | 3 | 7680 | 2360 | |
| 2328 | 2.0 | 0 | 4 | 3 | 6120 | 2550 | |
| 2364 | 1.5 | 0 | 1 | 5 | 3100 | 1760 | |
| 2654 | 1.0 | 1 | 4 | 3 | 4820 | 4820 | |
| 2712 | 2.0 | 0 | 3 | 3 | 3060 | 1280 | |
| 2761 | 2.0 | 0 | 0 | 3 | 6430 | 0 | |
| 2772 | 2.0 | 1 | 4 | 5 | 3850 | 1000 | |
| 2980 | 2.0 | 0 | 3 | 4 | 4430 | 1050 | |
| 3202 | 2.0 | 0 | 0 | 3 | 4560 | 1830 | |
| 3284 | 2.0 | 0 | 0 | 3 | 4250 | 0 | |
| 3729 | 1.0 | 0 | 2 | 4 | 4320 | 2730 | |
| 3781 | 2.0 | 0 | 2 | 4 | 3390 | 580 | |
| 3791 | 1.0 | 1 | 3 | 3 | 1840 | 1690 | |
| 3863 | 1.0 | 0 | 0 | 4 | 4040 | 0 | |

```
3959     2.0           0     1          3        4000          1010
4135     2.0           0     0          3        5180          1320
4261     2.0           0     0          4        4290             0
4273     2.0           0     0          3        4260          1520
4346     1.0           0     0          3        1550           640
4350     1.0           0     0          4        1180             0
4465     2.0           0     0          5        1300           410

      yr_built  yr_renovated                     street           city  \
1         1921             0             709 W Blaine St        Seattle
122       1999             0             26408 NE 70th St       Redmond
252       1910             0           1230 Warren Ave N        Seattle
404       2001             0   East Lake Sammamish Trail      Sammamish
581       1923             0            4437 55th Ave NE        Seattle
1132      2005             0               1010 6th St W       Kirkland
1543      1935             0         1404 Broadmoor Dr E        Seattle
1567      1999             0          5243 Forest Ave SE  Mercer Island
1571      1952          2007             3239 78th Pl NE         Medina
1600      1977          2005             9415 NE 27th St     Clyde Hill
1637      2000             0             3222 78th Pl NE         Medina
1858      1912          1994           309 W Prospect St        Seattle
2083      2001             0            17834 SE 40th Pl       Bellevue
2286      1940          2001            4442 95th Ave NE       Bellevue
2328      1965          2003           1291 NW Elford Dr        Seattle
2364      1905             0              173 37th Ave E        Seattle
2654      1983          2009         5044 Butterworth Rd  Mercer Island
2712      1993             0         5719 Seaview Ave NW        Seattle
2761      2001             0       1149-1199 91st Ave NE       Bellevue
2772      1989             0  11031 Champagne Point Rd NE       Kirkland
2980      1999             0               1223 7th Ave W        Seattle
3202      1940          1996           1239 Parkside Dr E        Seattle
3284      2004          2003              511 98th Ave NE       Bellevue
3729      1978          2000            1365 91st Ave NE     Clyde Hill
3781      1928             0           1323 Willard Ave W        Seattle
3791      1930          1993            7436 E Mercer Way  Mercer Island
3863      1954          1979           7623 Overlake Dr W         Medina
3959      2014             0            6410 E Mercer Way  Mercer Island
4135      2001             0             2205 78th Ave NE         Medina
4261      1985             0      1209 Evergreen Point Rd         Medina
4273      2006             0              9403 SE 33rd St  Mercer Island
4346      1956          2001             5426 40th Ave W        Seattle
4350      1992             0            12005 SE 219th Ct           Kent
4465      2003             0              2826 21st Ave W        Seattle

      statezip  price_per_sqft
1     WA 98119          653.15
122   WA 98053          168.39
```

```
252   WA 98109          515.30
404   WA 98075          800.00
581   WA 98105          482.08
1132  WA 98033          508.95
1543  WA 98112          518.71
1567  WA 98040          444.13
1571  WA 98039          620.77
1600  WA 98004          579.71
1637  WA 98039          668.47
1858  WA 98119          703.81
2083  WA 98008          610.86
2286  WA 98004          703.44
2328  WA 98177          333.10
2364  WA 98112          462.96
2654  WA 98040          484.23
2712  WA 98107          570.28
2761  WA 98004          698.13
2772  WA 98034          618.56
2980  WA 98119          437.96
3202  WA 98112          385.97
3284  WA 98004          541.18
3729  WA 98004          539.01
3781  WA 98119          579.35
3791  WA 98040          670.54
3863  WA 98039          562.17
3959  WA 98040          469.45
4135  WA 98039          378.15
4261  WA 98039          624.71
4273  WA 98040          401.56
4346  WA 98199         5889.95
4350  WA 98031        22533.90
4465  WA 98199         1497.37
```

### Predictive Model - Random Forest Model

[42]:
```
df['city'].unique()
```

[42]:
```
array(['Shoreline', 'Seattle', 'Kent', 'Bellevue', 'Redmond',
       'Maple Valley', 'North Bend', 'Lake Forest Park', 'Sammamish',
       'Auburn', 'Des Moines', 'Bothell', 'Federal Way', 'Kirkland',
       'Issaquah', 'Woodinville', 'Normandy Park', 'Fall City', 'Renton',
       'Carnation', 'Snoqualmie', 'Duvall', 'Burien', 'Covington',
       'Inglewood-Finn Hill', 'Kenmore', 'Newcastle', 'Mercer Island',
       'Black Diamond', 'Ravensdale', 'Clyde Hill', 'Algona', 'Skykomish',
       'Tukwila', 'Vashon', 'Yarrow Point', 'SeaTac', 'Medina',
       'Enumclaw', 'Snoqualmie Pass', 'Pacific', 'Beaux Arts Village',
       'Preston', 'Milton'], dtype=object)
```

[43]:
```
city_le = LabelEncoder()
df['city'] = city_le.fit_transform(df['city'])
```

[44]:
```
from sklearn.ensemble import RandomForestRegressor
X=df[['bedrooms', 'bathrooms','waterfront','condition','sqft_living','sqft_lot',
       'floors', 'view','sqft_above', 'city',
       'sqft_basement', 'yr_built', 'yr_renovated']]
y=df['price']

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.
 ↪2,random_state=42)
rf_model=RandomForestRegressor()
rf_model.fit(X_train,y_train)
rf_preds=rf_model.predict(X_test)
results_df=pd.DataFrame({'Condition':X_test["condition"],'bathrooms':
 ↪X_test["bathrooms"],'Actual':y_test,'predictions':rf_preds})
results_df = results_df[results_df['Actual']!=0]
results_df['abs_error']=abs(results_df['Actual']-results_df['predictions'])
results_df['abs_error_percentage']=results_df['abs_error']/results_df['Actual']
```

[45]:
```
results_df.describe()
```

[45]:

| | Condition | bathrooms | Actual | predictions | abs_error | \ |
|---|---|---|---|---|---|---|
| count | 911.000000 | 911.000000 | 9.110000e+02 | 9.110000e+02 | 9.110000e+02 | |
| mean | 3.440176 | 2.168222 | 5.861560e+05 | 5.386779e+05 | 1.735782e+05 | |
| std | 0.679762 | 0.777540 | 1.013750e+06 | 2.886379e+05 | 9.782922e+05 | |
| min | 1.000000 | 0.000000 | 8.330000e+04 | 1.647033e+05 | 3.839000e+01 | |
| 25% | 3.000000 | 1.750000 | 3.250000e+05 | 3.569001e+05 | 3.243859e+04 | |
| 50% | 3.000000 | 2.250000 | 4.584000e+05 | 4.778142e+05 | 7.575525e+04 | |
| 75% | 4.000000 | 2.500000 | 6.504400e+05 | 6.484630e+05 | 1.516331e+05 | |
| max | 5.000000 | 8.000000 | 2.659000e+07 | 4.873953e+06 | 2.635243e+07 | |

| | abs_error_percentage |
|---|---|
| count | 911.000000 |
| mean | 0.249822 |
| std | 0.298788 |
| min | 0.000108 |
| 25% | 0.070295 |
| 50% | 0.166572 |
| 75% | 0.312239 |
| max | 3.276287 |

Now add zipcode.

[46]:
```
df['statezip'].unique()
```

[46]:
```
array(['WA 98133', 'WA 98119', 'WA 98042', 'WA 98008', 'WA 98052',
       'WA 98115', 'WA 98038', 'WA 98045', 'WA 98155', 'WA 98105',
       'WA 98074', 'WA 98106', 'WA 98007', 'WA 98092', 'WA 98198',
       'WA 98006', 'WA 98102', 'WA 98011', 'WA 98125', 'WA 98003',
       'WA 98136', 'WA 98033', 'WA 98029', 'WA 98117', 'WA 98034',
       'WA 98072', 'WA 98023', 'WA 98107', 'WA 98166', 'WA 98116',
       'WA 98024', 'WA 98055', 'WA 98077', 'WA 98027', 'WA 98059',
       'WA 98075', 'WA 98014', 'WA 98065', 'WA 98199', 'WA 98053',
       'WA 98058', 'WA 98122', 'WA 98103', 'WA 98112', 'WA 98005',
       'WA 98118', 'WA 98177', 'WA 98004', 'WA 98019', 'WA 98144',
       'WA 98168', 'WA 98001', 'WA 98056', 'WA 98146', 'WA 98028',
       'WA 98148', 'WA 98057', 'WA 98040', 'WA 98010', 'WA 98051',
       'WA 98031', 'WA 98109', 'WA 98030', 'WA 98126', 'WA 98032',
       'WA 98178', 'WA 98288', 'WA 98108', 'WA 98070', 'WA 98188',
       'WA 98002', 'WA 98039', 'WA 98022', 'WA 98068', 'WA 98047',
       'WA 98050', 'WA 98354'], dtype=object)
```

[47]:
```
labels = ['WA 98133', 'WA 98119', 'WA 98042', 'WA 98008', 'WA 98052',
       'WA 98115', 'WA 98038', 'WA 98045', 'WA 98155', 'WA 98105',
       'WA 98074', 'WA 98106', 'WA 98007', 'WA 98092', 'WA 98198',
       'WA 98006', 'WA 98102', 'WA 98011', 'WA 98125', 'WA 98003',
       'WA 98136', 'WA 98033', 'WA 98029', 'WA 98117', 'WA 98034',
       'WA 98072', 'WA 98023', 'WA 98107', 'WA 98166', 'WA 98116',
       'WA 98024', 'WA 98055', 'WA 98077', 'WA 98027', 'WA 98059',
       'WA 98075', 'WA 98014', 'WA 98065', 'WA 98199', 'WA 98053',
       'WA 98058', 'WA 98122', 'WA 98103', 'WA 98112', 'WA 98005',
       'WA 98118', 'WA 98177', 'WA 98004', 'WA 98019', 'WA 98144',
       'WA 98168', 'WA 98001', 'WA 98056', 'WA 98146', 'WA 98028',
       'WA 98148', 'WA 98057', 'WA 98040', 'WA 98010', 'WA 98051',
       'WA 98031', 'WA 98109', 'WA 98030', 'WA 98126', 'WA 98032',
       'WA 98178', 'WA 98288', 'WA 98108', 'WA 98070', 'WA 98188',
       'WA 98002', 'WA 98039', 'WA 98022', 'WA 98068', 'WA 98047',
       'WA 98050', 'WA 98354']
z_le = LabelEncoder()
df['statezip'] = z_le.fit_transform(df['statezip'])
```

[48]:
```
from sklearn.ensemble import RandomForestRegressor
X=df[['bedrooms', 'bathrooms','waterfront','condition','sqft_living','sqft_lot',
       'floors', 'view','sqft_above', 'city','statezip',
       'sqft_basement', 'yr_built', 'yr_renovated']]
y=df['price']

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.
 ↪2,random_state=42)
rf_model=RandomForestRegressor()
rf_model.fit(X_train,y_train)
rf_preds=rf_model.predict(X_test)
```

```
results_df=pd.DataFrame({'Condition':X_test["condition"],'bathrooms':
  ↪X_test["bathrooms"],'Actual':y_test,'predictions':rf_preds})
results_df = results_df[results_df['Actual']!=0]
results_df['abs_error']=abs(results_df['Actual']-results_df['predictions'])
results_df['abs_error_percentage']=results_df['abs_error']/results_df['Actual']
```

[49]:
```
results_df.describe()
```

[49]:
```
         Condition   bathrooms        Actual   predictions     abs_error  \
count  911.000000  911.000000  9.110000e+02  9.110000e+02  9.110000e+02
mean     3.440176    2.168222  5.861560e+05  5.401922e+05  1.596169e+05
std      0.679762    0.777540  1.013750e+06  2.900564e+05  9.763713e+05
min      1.000000    0.000000  8.330000e+04  1.497730e+05  6.794700e+00
25%      3.000000    1.750000  3.250000e+05  3.548484e+05  3.055965e+04
50%      3.000000    2.250000  4.584000e+05  4.799285e+05  6.642711e+04
75%      4.000000    2.500000  6.504400e+05  6.448769e+05  1.336303e+05
max      5.000000    8.000000  2.659000e+07  4.491349e+06  2.635377e+07

       abs_error_percentage
count            911.000000
mean               0.224416
std                0.278612
min                0.000017
25%                0.071477
50%                0.146362
75%                0.269610
max                3.145780
```

Error Percentage is only ~15%!! Pretty accurate

[50]:
```
importances=rf_model.feature_importances_
indices=np.argsort(importances)[::-1]
sorted_features=X.columns[indices]
```

[51]:
```
sorted_features
```

[51]:
```
Index(['sqft_living', 'statezip', 'yr_built', 'city', 'sqft_above', 'sqft_lot',
       'sqft_basement', 'bathrooms', 'view', 'yr_renovated', 'bedrooms',
       'condition', 'floors', 'waterfront'],
      dtype='object')
```

Going to include top 6 variables for model only.

[52]:
```
from sklearn.ensemble import RandomForestRegressor
X=df[['sqft_living','sqft_lot', 'city','statezip',
     'sqft_above','yr_built']]
y=df['price']
```

```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.
 ↪2,random_state=42)
rf_model=RandomForestRegressor()
rf_model.fit(X_train,y_train)
rf_preds=rf_model.predict(X_test)
results_df=pd.DataFrame({'city':X_test["city"],'statezip':
 ↪X_test["statezip"],'Actual':y_test,'predictions':rf_preds})
results_df = results_df[results_df['Actual']!=0]
results_df['abs_error']=abs(results_df['Actual']-results_df['predictions'])
results_df['mean_abs_error'] =␣
 ↪abs(results_df['Actual']-results_df['predictions'])
results_df['abs_error_percentage']=results_df['abs_error']/results_df['Actual']
```

[53]:
```
results_df.describe()
```

[53]:

| | city | statezip | Actual | predictions | abs_error | \ |
|---|---|---|---|---|---|---|
| count | 911.000000 | 911.000000 | 9.110000e+02 | 9.110000e+02 | 9.110000e+02 | |
| mean | 25.237102 | 38.307355 | 5.861560e+05 | 5.423123e+05 | 1.609292e+05 | |
| std | 12.225032 | 20.652114 | 1.013750e+06 | 3.012114e+05 | 9.757195e+05 | |
| min | 0.000000 | 0.000000 | 8.330000e+04 | 1.505265e+05 | 1.114052e+02 | |
| 25% | 16.000000 | 22.000000 | 3.250000e+05 | 3.438275e+05 | 2.896914e+04 | |
| 50% | 32.000000 | 38.000000 | 4.584000e+05 | 4.779543e+05 | 6.704832e+04 | |
| 75% | 35.000000 | 56.000000 | 6.504400e+05 | 6.458309e+05 | 1.423912e+05 | |
| max | 43.000000 | 74.000000 | 2.659000e+07 | 4.554063e+06 | 2.634515e+07 | |

| | mean_abs_error | abs_error_percentage |
|---|---|---|
| count | 9.110000e+02 | 911.000000 |
| mean | 1.609292e+05 | 0.227412 |
| std | 9.757195e+05 | 0.292449 |
| min | 1.114052e+02 | 0.000437 |
| 25% | 2.896914e+04 | 0.062513 |
| 50% | 6.704832e+04 | 0.146696 |
| 75% | 1.423912e+05 | 0.276489 |
| max | 2.634515e+07 | 3.770379 |

Slightly more accurate!

[54]:
```
from sklearn.metrics import mean_absolute_error
mean_absolute_Error =␣
 ↪mean_absolute_error(results_df['Actual'],results_df['predictions'])
print("Mean Absolute Error: ", mean_absolute_Error)
```

```
Mean Absolute Error:  160929.2429611416
```

[55]:
```
results_df.describe()
```

[55]:
```
              city    statezip        Actual   predictions     abs_error  \
count   911.000000  911.000000  9.110000e+02  9.110000e+02  9.110000e+02
mean     25.237102   38.307355  5.861560e+05  5.423123e+05  1.609292e+05
std      12.225032   20.652114  1.013750e+06  3.012114e+05  9.757195e+05
min       0.000000    0.000000  8.330000e+04  1.505265e+05  1.114052e+02
25%      16.000000   22.000000  3.250000e+05  3.438275e+05  2.896914e+04
50%      32.000000   38.000000  4.584000e+05  4.779543e+05  6.704832e+04
75%      35.000000   56.000000  6.504400e+05  6.458309e+05  1.423912e+05
max      43.000000   74.000000  2.659000e+07  4.554063e+06  2.634515e+07

       mean_abs_error  abs_error_percentage
count    9.110000e+02            911.000000
mean     1.609292e+05              0.227412
std      9.757195e+05              0.292449
min      1.114052e+02              0.000437
25%      2.896914e+04              0.062513
50%      6.704832e+04              0.146696
75%      1.423912e+05              0.276489
max      2.634515e+07              3.770379
```

### Random Forest Model Applications

Now, want to see if we can use the random forest predictor model to predict prices for those houses that we set to the mean. The previous model tested and trained when some of the prices were equal to 0 and some prices were also outliers, meaning it may not be the most accurate. Now, I want to exclude when price = 0 and exclude outliers, to make it more accurate.

[56]:
```
notzero= df[df['price']!=0]
zero = df[df['price'] == 0]
notoutlierandnonzero = df[((z_scores<3)) & ((z_scores>-3)) & (df['price']!=0)]
from sklearn.ensemble import RandomForestRegressor
X=notoutlierandnonzero[['sqft_living','sqft_lot', 'city','statezip',
     'sqft_above','yr_built']]
y=notoutlierandnonzero['price']
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.
 ↪2,random_state=42)
rf_model=RandomForestRegressor()
rf_model.fit(X_train,y_train)
rf_preds=rf_model.predict(X_test)
results_df=pd.DataFrame({'city':X_test["city"],'statezip':
 ↪X_test["statezip"],'Actual':y_test,'predictions':rf_preds})
results_df = results_df[results_df['Actual']!=0]
results_df['abs_error']=abs(results_df['Actual']-results_df['predictions'])
results_df['abs_error_percentage']=results_df['abs_error']/results_df['Actual']
```

Model has now been created based on houses that actually have a price- median absolute error percentage is more accurate now that price=0 has been excluded. Now putting the model to use to predict the prices where price is 0 and where the price is an outlier to make it more accurate.

[57]:
```
results_df.describe()
```

[57]:

| | city | statezip | Actual | predictions | abs_error | \ |
|---|---|---|---|---|---|---|
| count | 904.000000 | 904.000000 | 9.040000e+02 | 9.040000e+02 | 904.000000 | |
| mean | 25.838496 | 38.871681 | 5.302167e+05 | 5.386769e+05 | 103022.925312 | |
| std | 12.014407 | 20.574827 | 2.864133e+05 | 2.575750e+05 | 126995.384760 | |
| min | 0.000000 | 0.000000 | 1.000000e+05 | 1.465054e+05 | 50.636700 | |
| 25% | 17.000000 | 21.000000 | 3.278750e+05 | 3.547714e+05 | 26143.486250 | |
| 50% | 32.000000 | 41.500000 | 4.500000e+05 | 4.802917e+05 | 60523.269150 | |
| 75% | 35.000000 | 56.000000 | 6.555000e+05 | 6.701145e+05 | 123963.064300 | |
| max | 42.000000 | 74.000000 | 2.000000e+06 | 1.904445e+06 | 860699.120000 | |

| | abs_error_percentage |
|---|---|
| count | 904.000000 |
| mean | 0.206322 |
| std | 0.255941 |
| min | 0.000058 |
| 25% | 0.057686 |
| 50% | 0.131254 |
| 75% | 0.264692 |
| max | 3.458614 |

[58]:
```
X_notoutlierandnonzero =␣
↪notoutlierandnonzero[['sqft_living','sqft_lot','city','statezip','sqft_above','yr_built']]
notoutlierandnonzero_preds = rf_model.predict(X_notoutlierandnonzero)
results_notoutlierandnonzero = pd.DataFrame({'sqft_living':
↪X_notoutlierandnonzero["sqft_living"],'statezip':
↪X_notoutlierandnonzero["statezip"], 'city' : X_notoutlierandnonzero["city"],␣
↪'sqft_lot': X_notoutlierandnonzero["sqft_lot"], 'sqft_above' :
↪X_notoutlierandnonzero["sqft_above"], 'yr_built':␣
↪X_notoutlierandnonzero["yr_built"], 'predictions':
↪notoutlierandnonzero_preds})
```

[59]:
```
results_notoutlierandnonzero
```

[59]:

| | sqft_living | statezip | city | sqft_lot | sqft_above | yr_built | predictions |
|---|---|---|---|---|---|---|---|
| 0 | 1340 | 62 | 36 | 7912 | 1340 | 1955 | 321194.2857 |
| 2 | 1930 | 26 | 18 | 11947 | 1930 | 1966 | 329333.7296 |
| 3 | 2000 | 7 | 3 | 8030 | 1000 | 1963 | 447185.0000 |
| 4 | 1940 | 31 | 31 | 10500 | 1140 | 1976 | 541014.0000 |
| 5 | 880 | 54 | 35 | 6380 | 880 | 1938 | 460606.7085 |
| … | … | … | … | … | … | … | … |
| 4595 | 1510 | 62 | 35 | 6360 | 1510 | 1954 | 331590.8000 |
| 4596 | 1460 | 6 | 3 | 7573 | 1460 | 1983 | 462062.0000 |
| 4597 | 3010 | 37 | 32 | 7014 | 3010 | 2009 | 485261.7795 |
| 4598 | 2090 | 71 | 35 | 6630 | 1070 | 1974 | 389029.2500 |
| 4599 | 1490 | 26 | 9 | 8102 | 1490 | 1990 | 209696.0000 |

```
[4517 rows x 7 columns]
```

[60]:
```
from sklearn.metrics import mean_absolute_error
mean_absolute_Error =␣
 ↪mean_absolute_error(results_df['Actual'],results_df['predictions'])
print("Mean Absolute Error: $", mean_absolute_Error)
```

```
Mean Absolute Error: $ 103022.92531235248
```

[61]:
```
def predictions(sqft_living,sqft_lot,city,statezip,sqft_above,yr_built):
    city = city_le.transform([city])[0]
    statezip = z_le.transform([statezip])[0]
    house_price = [[sqft_living,sqft_lot,city,statezip,sqft_above,yr_built]]
    rf_preds=rf_model.predict(house_price)
    results_df=pd.DataFrame({'predictions':rf_preds})
    return results_df
```

[62]:
```
predictions(3000,790,'Bellevue','WA 98006',400,1988)
```

```
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-
packages/sklearn/utils/validation.py:2749: UserWarning: X does not have valid
feature names, but RandomForestRegressor was fitted with feature names
  warnings.warn(
```

[62]:

| | predictions |
|---|---|
| 0 | 804138.0974 |

[63]:
```
housing.head()
```

[63]:

| | date | price | bedrooms | bathrooms | sqft_living | sqft_lot | floors | \ |
|---|---|---|---|---|---|---|---|---|
| 0 | 2014-05-02 | 313000.0 | 3.0 | 1.50 | 1340 | 7912 | 1.5 | |
| 1 | 2014-05-02 | 2384000.0 | 5.0 | 2.50 | 3650 | 9050 | 2.0 | |
| 2 | 2014-05-02 | 342000.0 | 3.0 | 2.00 | 1930 | 11947 | 1.0 | |
| 3 | 2014-05-02 | 420000.0 | 3.0 | 2.25 | 2000 | 8030 | 1.0 | |
| 4 | 2014-05-02 | 550000.0 | 4.0 | 2.50 | 1940 | 10500 | 1.0 | |

| | waterfront | view | condition | sqft_above | sqft_basement | yr_built | \ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 3 | 1340 | 0 | 1955 | |
| 1 | 0 | 4 | 5 | 3370 | 280 | 1921 | |
| 2 | 0 | 0 | 4 | 1930 | 0 | 1966 | |
| 3 | 0 | 0 | 4 | 1000 | 1000 | 1963 | |
| 4 | 0 | 0 | 4 | 1140 | 800 | 1976 | |

| | yr_renovated | street | city | statezip | price_per_sqft |
|---|---|---|---|---|---|
| 0 | 2005 | 18810 Densmore Ave N | Shoreline | WA 98133 | 233.58 |
| 1 | 0 | 709 W Blaine St | Seattle | WA 98119 | 653.15 |

```
2             0  26206-26214 143rd Ave SE       Kent  WA 98042          177.20
3             0            857 170th Pl NE   Bellevue  WA 98008          210.00
4          1992          9105 170th Ave NE    Redmond  WA 98052          283.51
```

[64]:
```
housing.tail()
```

[64]:
```
            date      price  bedrooms  bathrooms  sqft_living  sqft_lot  \
4595  2014-07-09  308166.67       3.0       1.75         1510      6360
4596  2014-07-09  534333.33       3.0       2.50         1460      7573
4597  2014-07-09  416904.17       3.0       2.50         3010      7014
4598  2014-07-10  203400.00       4.0       2.00         2090      6630
4599  2014-07-10  220600.00       3.0       2.50         1490      8102

      floors  waterfront  view  condition  sqft_above  sqft_basement  \
4595     1.0           0     0          4        1510              0
4596     2.0           0     0          3        1460              0
4597     2.0           0     0          3        3010              0
4598     1.0           0     0          3        1070           1020
4599     2.0           0     0          4        1490              0

      yr_built  yr_renovated             street       city  statezip  \
4595      1954          1979     501 N 143rd St    Seattle  WA 98133
4596      1983          2009   14855 SE 10th Pl   Bellevue  WA 98007
4597      2009             0   759 Ilwaco Pl NE     Renton  WA 98059
4598      1974             0  5148 S Creston St    Seattle  WA 98178
4599      1990             0  18717 SE 258th St  Covington  WA 98042

      price_per_sqft
4595          204.08
4596          365.98
4597          138.51
4598           97.32
4599          148.05
```

[65]:
```
predictions(1340,7912,'Shoreline','WA 98133', 1340, 1955)
```

```
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-
packages/sklearn/utils/validation.py:2749: UserWarning: X does not have valid
feature names, but RandomForestRegressor was fitted with feature names
  warnings.warn(
```

[65]:
```
   predictions
0  321194.2857
```

Very close to the actual price of this listing in the table!! Only about 10,000 dollars off.

[66]:
```
predictions(3650,9050,'Seattle','WA 98119',3370, 1921)
```

```
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-
packages/sklearn/utils/validation.py:2749: UserWarning: X does not have valid
feature names, but RandomForestRegressor was fitted with feature names
  warnings.warn(
```

[66]:

| | predictions |
|---|---|
| 0 | 1.147454e+06 |

[67]:
```
predictions(1510, 6360, 'Seattle', 'WA 98133', 1510, 1954)
```

```
/opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-
packages/sklearn/utils/validation.py:2749: UserWarning: X does not have valid
feature names, but RandomForestRegressor was fitted with feature names
  warnings.warn(
```

[67]:

| | predictions |
|---|---|
| 0 | 331590.8 |

Linear Regression

[68]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X=notoutlierandnonzero[['sqft_living','sqft_lot', 'city','statezip',
     'sqft_above','yr_built']]
y=notoutlierandnonzero['price']

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.
 ↪2,random_state=42)
linear = LinearRegression()
linear.fit(X_train,y_train)
```

[68]:
```
LinearRegression()
```

[69]:
```
y_preds=linear.predict(X_test)
results_df=pd.DataFrame({'Actual':y_test,'predictions':y_preds})
results_df['abs_error']=abs(results_df['Actual']-results_df['predictions'])
results_df['abs_error_percentage']=results_df['abs_error']/results_df['Actual']
results_df.describe()
```

[69]:

| | Actual | predictions | abs_error | abs_error_percentage |
|---|---|---|---|---|
| count | 9.040000e+02 | 9.040000e+02 | 904.000000 | 904.000000 |
| mean | 5.302167e+05 | 5.356946e+05 | 147514.572819 | 0.324076 |
| std | 2.864133e+05 | 2.158009e+05 | 134064.770205 | 0.306504 |
| min | 1.000000e+05 | 5.270500e+04 | 516.475022 | 0.001179 |
| 25% | 3.278750e+05 | 3.799237e+05 | 54734.900695 | 0.109386 |
| 50% | 4.500000e+05 | 4.970794e+05 | 114354.536642 | 0.241184 |
| 75% | 6.555000e+05 | 6.524198e+05 | 200484.997996 | 0.435715 |

```
max    2.000000e+06  1.499467e+06  983713.728118              2.709060
```

Linear Model is much less accurate than random forest model (to be expected)

[70]:
```
lastdf = pd.DataFrame({
    'model': ['Linear Regression', 'Random Forest'],
    'median_abs_error_percentage':[24.1,13.5]
})
lastdf
```

[70]:

| | model | median_abs_error_percentage |
|---|---|---|
| 0 | Linear Regression | 24.1 |
| 1 | Random Forest | 13.5 |

[71]:
```
plt.figure(figsize=(7,5))
sns.set_theme(style="ticks")
sns.barplot(y = 'median_abs_error_percentage', x = "model", data = lastdf)
plt.ylim(0,100)
plt.title('Results of Different Model Types in Predicting House Prices in␣
 ↪Washington')
plt.xlabel('Model Type')
plt.ylabel('median absolute error percentage')
plt.show()
```

